

AP is a sequence of numbers in which the difference between consecutive terms is constant.

1) 7 11 15 19 23 27

$+4$ $+4$ $+4$ $+4$ $+4$

2) 101 $\xrightarrow{+5}$ 106 $\xrightarrow{+5}$ 111 $\xrightarrow{+5}$ 116 $\xrightarrow{+5}$ 121 $\xrightarrow{+5}$ 126

3) 74 84 94 104 114 124

$+10$ $+10$ $+10$ $+10$ $+10$

Same gap

and
numbers
are

increasing

This also works when numbers are decreasing with the same gap.

4) 47 45 43 41 39 37 35

-2 -2 -2 -2 -2 -2

5) $100 \xrightarrow{-9} 91 \xrightarrow{-9} 82 \xrightarrow{-9} 73 \xrightarrow{-9} 64 \xrightarrow{-9} 55 \xrightarrow{-9} 46$

6) $39 \xrightarrow{-1} 38 \xrightarrow{-1} 37 \xrightarrow{-1} 36 \xrightarrow{-1} 35 \xrightarrow{-1} 34 \xrightarrow{-1} 33$

Basic form of A.P:-

- a = first term
- d = common difference

$$a, \quad a+d, \quad a+2d, \quad a+3d, \quad a+4d, \quad \dots$$

$\xrightarrow{+d} \quad \xrightarrow{+d} \quad \xrightarrow{+d} \quad \xrightarrow{+d}$

- * When d is +ve : Sequence increases
- * When d is -ve : Sequence decreases

$$\begin{array}{ccccccccc}
 \boxed{a} & \boxed{a+d} & \boxed{a+2d} & \boxed{a+3d} & \boxed{a+4d} & , \dots & \boxed{a+(n-1)d} \\
 \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} & & n^{\text{th}} \text{ term}
 \end{array}$$

$$A_n = a + (n-1)d$$

$$A_1 = a + 0d = a$$

$$A_2 = a + d$$

$$A_3 = a + 2d$$

$$A_4 = a + 3d$$

$$A_5 = a + 4d$$

$$A_6 = a + 5d$$

$$A_7 = a + 6d$$

⋮

$$A_n = a + (n-1)d$$

$a, a+d, a+2d, a+3d, a+4d, \dots, a+(n-1)d$

• $\text{Average} = \frac{\text{First term} + \text{Last term}}{2}$

$\left\{ \text{Average} = \frac{\text{Sum}}{n} \right\}$

• $\text{Sum} = \text{Average} \times n$

$\text{Sum} = \frac{(\text{First} + \text{Last})}{2} \times n$

* $\text{Sum} = \frac{n}{2} \{ \text{First} + \text{Last} \}$

$\Rightarrow \text{Sum} = \frac{n}{2} \{ (a) + (a + (n-1)d) \}$

$\text{Sum} = \frac{n}{2} \{ a + a + (n-1)d \}$

* $\text{Sum} = \frac{n}{2} \{ 2a + (n-1)d \}$

where,

n = number of terms

a = first term

d = common difference

S = Sum

Q1: The first term of an AP is 7 and the common difference is 4. What is the 9th term?

एक AP का पहला पद 7 है और सार्व अंतर 4 है। इसका 9वाँ पद क्या है?

(A) 39

(B) 35

(C) 43

(D) 47

(E) 51

$$A_n = a + (n-1)d$$

$$A_9 = a + (9-1)d$$

$$A_9 = a + 8d$$

$$A_9 = 7 + 8(4)$$

$$A_9 = 7 + 32$$

$$A_9 = 39$$

$$\left\{ \begin{array}{l} \text{Given,} \\ a = 7 \\ d = 4 \end{array} \right\}$$

Q2: What is the 15th term of the AP 11, 16, 21, ...?

AP 11, 16, 21, ... का 15वाँ पद क्या है?

(A) 71

(B) 76

(C) 81

(D) 86

(E) 91

$$A_n = a + (n-1)d$$

$$A_{15} = a + (15-1)d$$

$$A_{15} = a + 14d$$

$$A_{15} = 11 + 14(5)$$

$$A_{15} = 11 + 70$$

$$A_{15} = 81$$

$$\left\{ \begin{array}{l} \text{where,} \\ a = 11 \\ d = 16 - 11 = 5 \\ 21 - 16 = 5 \end{array} \right\}$$

Q3: The first term of an AP is 42 and the common difference is -3. Find the 10th term.

एक AP का पहला पद 42 है और सार्व अंतर -3 है। इसका 10वाँ पद ज्ञात कीजिए।

(A) 6

(B) 9

(C) 12

(D) 15

(E) 18

$$\text{Given, } a = 42$$

$$d = -3$$

$$A_{10} = a + 9d$$

$$A_{10} = 42 + 9(-3)$$

$$A_{10} = 42 - 27$$

$$A_{10} = 15$$

Q4: In an AP, the 12th term is 46 and the common difference is 3. Find the first term.

एक AP में, 12वाँ पद 46 है और सार्व अंतर 3 है। पहला पद ज्ञात कीजिए।

(A) 11

(B) 13

(C) 15

(D) 17

(E) 19

$$A_{12} = a + 11d$$

$$a + 11d = 46$$

$$a + 11(3) = 46$$

$$a = 46 - 33$$

$$\boxed{a = 13}$$

$$\left\{ \begin{array}{l} \text{Given,} \\ A_{12} = 46 \\ d = 3 \end{array} \right\}$$

Q5: In an AP, the 4th term is 20 and the 12th term is 52. Find the 20th term.

एक AP में, चौथा पद 20 है और 12वाँ पद 52 है। 20वाँ पद ज्ञात कीजिए।

(A) 76

(B) 80

(C) 84

(D) 88

(E) 92

$$A_4 = a + 3d \rightarrow 20$$

$$\downarrow + 8d$$

$$\downarrow + 32$$

$$A_{12} = a + 11d \rightarrow 52$$

$$\downarrow + 8d$$

$$\downarrow + 32$$

$$\left\{ \begin{array}{l} 8d = 32 \\ d = 4 \end{array} \right\}$$

$$A_{20} = a + 19d \rightarrow \boxed{84}$$

Q6: In an AP, the 5th term is 23 and the 8th term is 38. Find the common difference.

एक AP में, 5वाँ पद 23 है और 8वाँ पद 38 है। सार्व अंतर ज्ञात कीजिए।

(A) 4

(B) 5

(C) 6

(D) 7

(E) 8

$$A_5 = a + 4d \rightarrow 23$$

$$\downarrow + 3d$$

$$\downarrow + 15$$

$$A_8 = a + 7d \rightarrow 38$$

$$\# \quad 3d = 15$$

$$\boxed{d = 5}$$

Q7: Which term of the AP 9, 15, 21, ... is 129?

AP 9, 15, 21, ... का कौन-सा पद 129 है?

(A) 20

☒ (B) 21

(C) 22

(D) 23

(E) 24

AP $\rightarrow$ 9, 15, 21, 27, 33, 39, ..., 129
 A_1 A_2 A_3 A_4 A_5 A_6 A_n

$$\left\{ \begin{array}{l} d = 15 - 9 \Rightarrow 6 \\ 21 - 15 \Rightarrow 6 \end{array} \right\}$$

$$A_n = a + (n-1)d$$

$$129 = 9 + (n-1)6$$

$$20 \quad \cancel{120} = (n-1)6$$

$$n-1 = 20$$

$$\boxed{n = 21}$$

Q8: The first term of an AP is 50 and the common difference is -4. Which term is 10?

एक AP का पहला पद 50 है और सार्व अंतर -4 है। इसका कौन-सा पद 10 है?

(A) 7

(B) 8

(C) 9

(D) 10

☒ (E) 11

$$a = 50$$

$$d = -4$$

AP $\rightarrow$ 50, 46, 42, 38, 34, ..., 10
 A_1 A_2 A_3 A_4 A_5 A_n

$$A_n = a + (n-1)d$$

$$10 = 50 + (n-1)(-4)$$

$$10 - 50 = (n-1)(-4)$$

$$\frac{-40}{-4} = n-1$$

$$n-1 = 10$$

$$\boxed{n = 11}$$

Geometric Progression :-

GP is a sequence of numbers where each term is multiplied by the same number to get the next term.

$$1) \quad 5, 10, 20, 40, 80, 160, 320, 640$$

$\xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2} \quad \xrightarrow{\times 2}$

$$2) \quad 80, 120, 180, 270, 405, 607.5$$

$\xrightarrow{\times \frac{3}{2}} \quad \xrightarrow{\times \frac{3}{2}} \quad \xrightarrow{\times \frac{3}{2}} \quad \xrightarrow{\times \frac{3}{2}} \quad \xrightarrow{\times \frac{3}{2}}$

$$3) \quad 729, 243, 81, 27, 9, 3, 1$$

$\xrightarrow{\times \frac{1}{3}} \quad \xrightarrow{\times \frac{1}{3}} \quad \xrightarrow{\times \frac{1}{3}} \quad \xrightarrow{\times \frac{1}{3}} \quad \xrightarrow{\times \frac{1}{3}} \quad \xrightarrow{\times \frac{1}{3}}$

Basic form of GP :-

a = first term

r = common ratio

$$G_1, G_2, G_3, G_4, G_5, G_6, \dots, G_n$$

$$a, ar, ar^2, ar^3, ar^4, ar^5, \dots, ar^{n-1}$$

$\xrightarrow{\times r} \quad \xrightarrow{\times r} \quad \xrightarrow{\times r} \quad \xrightarrow{\times r} \quad \xrightarrow{\times r}$

$$G_n = ar^{n-1}$$

where,

a = first term

r = common ratio

Q9: In the GP 3, 6, 12, 24, ..., what is the next term?

GP 3, 6, 12, 24, ... में, अगला पद क्या है?

(A) 36

(B) 42

(C) 48

(D) 54

(E) 60

$$a = 3$$

GP $\rightarrow$ 3, 6, 12, 24, 48

$\underbrace{\quad \times 2 \quad}$
 $\underbrace{\quad \times 2 \quad}$
 $\underbrace{\quad \times 2 \quad}$
 $\underbrace{\quad \times 2 \quad}$

Q10: If the first term of a GP is 5 and the common ratio is 3, what is the 5th term?

यदि किसी GP का पहला पद 5 है और सार्व अनुपात 3 है, तो उसका 5वाँ पद क्या होगा?

(A) 81

(B) 135

(C) 243

(D) 405

(E) 540

$$a = 5$$

$$r = 3$$

$$T_5 = ar^{(5-1)}$$

$$T_5 = ar^4$$

$$= (5) \times (3)^4$$

$$= 5 \times 81$$

$$= 405$$

Q11: The 4th term of a GP is 16 and the common ratio is $\frac{1}{2}$. What is the first term?

एक GP का चौथा पद 16 है और सार्व अनुपात $\frac{1}{2}$ है। इसका पहला पद क्या है?

(A) 64

(B) 96

(C) 128

(D) 144

(E) 160

$$T_4 = ar^{(4-1)}$$

$$T_4 = ar^3$$

$$16 = a \left[\frac{1}{2} \right]^3$$

$$16 = a \times \frac{1}{8}$$

$$a = 128$$

$$\left\{ \begin{array}{l} \text{Given,} \\ T_4 = 16 \\ r = \frac{1}{2} \end{array} \right\}$$

Q12: In a GP, the first term is 2 and the common ratio is 3. Which term is 486?

एक GP में, पहला पद 2 है और सार्व अनुपात 3 है। कौन-सा पद 486 है?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

$$a = 2$$

$$r = 3$$

$$G_n = ar^{(n-1)}$$

$$486 = (2)(3)^{n-1}$$

$$\frac{486}{2} = (3)^{n-1}$$

$$243 = 3^{(n-1)}$$

$$3^5 = 3^{n-1} \Rightarrow n-1 = 5$$

$$\boxed{n = 6}$$

Q13: The 2nd term of a GP is 10 and the 5th term is 80. What is the common ratio?

एक GP का दूसरा पद 10 है और पाँचवाँ पद 80 है। इसका सार्व अनुपात क्या है?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Given, $G_2 = 10$

$$G_5 = 80$$

$$\frac{ar^1}{ar^4} = \frac{10}{80}$$

{ Dividing both terms }

$$\frac{ar}{ar^4} = \frac{1}{8}$$

$$\frac{1}{r^3} = \frac{1}{8}$$

$$\frac{1}{r^3} = \frac{1}{2^3}$$

$$\Rightarrow r^3 = 2^3$$

$$\boxed{r = 2}$$

Q14: In the GP 81, 27, 9, 3, ..., what is the 6th term?

GP 81, 27, 9, 3, ... में, छठा पद क्या है?

(A) $1/9$

☒ (B) $1/3$

(C) 1

(D) 3

(E) 9

$$81 \xrightarrow{\times \frac{1}{3}} 27 \xrightarrow{\times \frac{1}{3}} 9 \xrightarrow{\times \frac{1}{3}} 3, \dots$$

$$\begin{cases} a = 81 \\ r = \frac{1}{3} \end{cases}$$

$$G_6 = ar^5$$

$$G_6 = 81 \left[\frac{1}{3} \right]^5$$

$$G_6 = \frac{81 \times 1}{243} \Rightarrow \boxed{G_6 = \frac{1}{3}}$$

Q15: The 3rd term of a GP is 20 and the common ratio is 2. What is the 6th term?

एक GP का तीसरा पद 20 है और सार्व अनुपात 2 है। इसका छठा पद क्या है?

(A) 80

(B) 100

(C) 120

(D) 140

☒ (E) 160

$$G_3 \Rightarrow ar^2 = 20$$

$\Rightarrow$

$$G_6 = ar^{6-1}$$

$$a(2)^2 = 20$$

$$G_6 = ar^5$$

$$a = \frac{20}{4}$$

$$G_6 = (5)(2)^5$$

$$\boxed{a = 5}$$

$$G_6 = 5 \times 32$$

$$\boxed{G_6 = 160}$$

Method 2 : $G_3 = ar^2 = 20$

$$ar^2 \xrightarrow{\times r^3} ar^5$$

$$20 \xrightarrow[\times 8]{\times (2)^3} \boxed{160}$$